

MercaMiner v0.1

Step-by-Step Running Guide

Build, configure, start, monitor, benchmark, and stop the Mercatura reference CPU miner

Target platform: Linux x86_64
MercaHash V1 | Solo JSON-RPC mining
Guide version: 23 September 2026

Important: This guide uses placeholder values such as `YOUR_TESTNET_MERCATURA_ADDRESS`. Replace the placeholder text with the real value. Do not type angle brackets around placeholders in a shell command.

1. What you need before mining

MercaMiner mines directly against a running Mercatura Core node. Before starting the miner, make sure the following are available:

- A built or installed copy of Mercatura Core.
- A built copy of MercaMiner.
- A Mercatura payout address for the selected network.
- Enough free RAM for the selected number of MercaHash workers.
- RPC access to Mercatura Core, either through the default local cookie or explicit RPC settings.

Verified v0.1 status: The current miner has completed local regtest and testnet validation. A live testnet run completed 50 accepted blocks with 0 stale and 0 rejected blocks before clean shutdown.

2. Build MercaMiner

Step 1 - Open the MercaMiner project directory.

```
cd ~/projects/mercaminer
```

Step 2 - Install build dependencies if they are not already installed.

```
sudo apt update
sudo apt install -y build-essential cmake libcurl4-openssl-dev nlohmann-json3-dev
```

Step 3 - Configure and build MercaMiner.

```
cmake -S . -B build
cmake --build build -j"${nproc}"
```

Step 4 - Run the test suite.

```
ctest --test-dir build --output-on-failure
```

A healthy v0.1 build should complete the MercaMiner test suite without failures. The primary executable is:

```
./build/mercaminer
```

3. Start or verify Mercatura Core

MercaMiner needs a running Mercatura Core node on the same network you intend to mine. The examples below assume Mercatura Core was built from source under ~/projects/mercatura.

Testnet

```
cd ~/projects/mercatura
./build/bin/mercaturad -testnet -server=1 -daemon
./build/bin/mercatura-cli -testnet getblockchaininfo
```

Mainnet

```
cd ~/projects/mercatura
./build/bin/mercaturad -server=1 -daemon
./build/bin/mercatura-cli getblockchaininfo
```

Regtest

```
cd ~/projects/mercatura
./build/bin/mercaturad -regtest -server=1 -daemon
./build/bin/mercatura-cli -regtest getblockchaininfo
```

Custom data directory: If Mercatura Core is started with `-datadir=PATH`, MercaMiner cannot use the normal default cookie path automatically. Use explicit `--rpc-url` and `--cookie-file` settings as shown later in this guide.

4. Obtain a payout address

Use a Mercatura wallet address that belongs to the same network being mined. The block reward is paid to this address.

You can copy a receive address from Mercatura Core Qt. If the appropriate wallet is loaded in Mercatura Core, the CLI form is typically:

```
cd ~/projects/mercatura
./build/bin/mercatura-cli -testnet getnewaddress
```

For mainnet, omit `-testnet`. For regtest, replace it with `-regtest`. If multiple wallets are in use, select the intended wallet using normal Mercatura Core wallet RPC handling.

Network match: A mainnet address must be used with mainnet, a testnet address with testnet, and a regtest address with regtest. MercaMiner validates network identity before mining.

5. Quick start - command-line mining

If Mercatura Core uses its normal data directory, MercaMiner can automatically discover the local RPC endpoint and authentication cookie.

Step 1 - Change to the MercaMiner directory.

```
cd ~/projects/mercaminer
```

Step 2 - Start the miner. Replace the payout placeholder with your real address.

```
./build/mercaminer \
--network test \
--payout-address YOUR_TESTNET_MERCATURA_ADDRESS \
--threads 6 \
--block-limit 0 \
--report-interval 30
```

This example mines testnet continuously with 6 workers and prints a status line about every 30 seconds.

Supported direct-mining network names

Network	CLI value	Direct PoW mining
Mainnet	main	Yes
Testnet	test	Yes
Regtest	regtest	Yes
Signet	signet	No - identity recognized, mining blocked

Bitcoin Testnet4 is not supported by MercaMiner v0.1.

6. Recommended configuration-file setup

For normal day-to-day use, a configuration file is more convenient than retyping the command line.

Step 1 - Create the configuration directory.

```
mkdir -p ~/.mercaminer
```

Step 2 - Create ~/.mercaminer/mercaminer.conf with the following contents, replacing the payout address.

```
network=test
payout_address=YOUR_TESTNET_MERCATURA_ADDRESS
threads=6
block_limit=0
report_interval=30
```

Step 3 - Start MercaMiner with no arguments.

```
cd ~/projects/mercaminer
./build/mercaminer
```

The default configuration path is `~/mercaminer/mercaminer.conf`. An alternate file can be selected with:

```
./build/mercaminer --config /path/to/mercaminer.conf
```

Command-line options override configuration-file values.

7. Configuration settings explained

Setting	Meaning	Typical value
network	Mercatura network to mine	test
payout_address	Address receiving mined block rewards	Your network-matched address
threads	Number of MercaHash worker threads	6 on the validated Ryzen 5 4600G example
block_limit	Accepted blocks before automatic stop; 0 = continuous	0
report_interval	Status interval in seconds; 0 = disabled	30
rpc_url	Explicit node RPC URL; optional with default local setup	http://127.0.0.1:27775
cookie_file	Explicit RPC cookie path; must accompany rpc_url	Path to .cookie

8. Using a custom Mercatura Core data directory

If the node uses a custom `-datadir`, supply both the RPC URL and cookie file. For example, a testnet node using a custom directory might be launched with:

```
cd ~/projects/mercatura
./build/bin/mercaturad \
  -testnet \
  -datadir=/path/to/mercatura-testnet \
  -server=1 \
  -daemon
```

Then start MercaMiner with the matching RPC settings:

```
cd ~/projects/mercaminer
./build/mercaminer \
  --network test \
  --payout-address YOUR_TESTNET_MERCATURA_ADDRESS \
  --threads 6 \
  --block-limit 0 \
  --report-interval 30 \
  --rpc-url http://127.0.0.1:27775 \
  --cookie-file /path/to/mercatura-testnet/testnet/.cookie
```

Pairing rule: `rpc_url` and `cookie_file` must be supplied together. Supplying only one is a configuration error.

9. Default local RPC locations

Network	Default RPC port	Default cookie file
main	27776	<code>~/.mercatura/.cookie</code>
test	27775	<code>~/.mercatura/testnet/.cookie</code>
regtest	27773	<code>~/.mercatura/regtest/.cookie</code>

MercaMiner can reload RPC credentials after Mercatura Core rotates its cookie and can wait for RPC availability during normal startup or recovery.

10. Choose a worker count

MercaHash V1 uses a private 128 MiB mutable scratchpad per worker. More workers increase memory use and do not necessarily scale linearly.

Workers	MercaHash scratchpad memory
1	128 MiB
2	256 MiB
4	512 MiB
6	768 MiB
8	1 GiB
12	1.5 GiB

Validated example: On the Ryzen 5 4600G development system, 6 workers produced about 32.7 H/s in sustained benchmarking and the live testnet run stabilized around 70-71 C. This is a machine-specific reference, not a universal target.

If `threads` is omitted, MercaMiner automatically selects a worker count based on detected hardware, capped at 8 workers by default.

11. Benchmark before long-term mining

MercaMiner includes a sustained MercaHash benchmark. Use it to compare thread counts on the actual machine.

```
cd ~/projects/mercaminer
./build/mercaminer_benchmark 6 180
```

The first argument is the worker count and the second is the measurement duration in seconds. Longer runs are more useful than very short tests because the reported average needs time to converge.

12. Understand the live output

A normal periodic status line resembles:

```
[status] height=51 workers=6 hashrate=31.80 H/s session_hashes=269042 accepted=50 stale=0
rejected=0 uptime=02:19:30
```

- **height** - The next block height MercaMiner is currently trying to mine.
- **workers** - Active MercaHash worker count.
- **hashrate** - Current reported session hashrate.
- **session_hashes** - Total MercaHash attempts checked during the session.
- **accepted** - Blocks accepted by Mercatura Core.
- **stale** - Valid/inconclusive work that lost a tip race or became outdated.
- **rejected** - Genuine block-submission validation failures.
- **uptime** - Current miner session duration.

When a block is accepted, MercaMiner also prints the height, nonce, extranonce, MercaHash, block hash, active workers, and hash counters.

13. Stop the miner cleanly

Press Ctrl+C once in the terminal running MercaMiner. The miner requests worker shutdown and prints a final session summary.

```
Ctrl+C
```

A normal clean shutdown ends with totals for accepted blocks, stale work, rejected blocks, and session hashes checked.

14. Built-in safety checks

Before mining, MercaMiner checks the connected node and template. Important protections include:

- RPC chain identity must match the requested network.
- The live genesis block must match MercaMiner's expected Mercatura genesis for that network.
- The selected network must support direct proof-of-work mining.
- The block-template target must not exceed Mercatura's configured proof-of-work limit.

15. Troubleshooting

Bash says "No such file or directory" when using a placeholder address

Do not enter placeholders with angle brackets such as <YOUR_ADDRESS>. In a shell, < is input redirection. Use the real address directly, or use plain placeholder text such as YOUR_TESTNET_MERCATURA_ADDRESS when writing examples.

MercaMiner cannot find the RPC cookie

Confirm Mercatura Core is running on the same network. If Core uses a custom `-datadir`, supply both `--rpc-url` and `--cookie-file` explicitly.

"rpc_url and cookie_file must be specified together"

Provide both settings or remove both and let MercaMiner use automatic local RPC discovery.

Genesis hash mismatch

The connected node does not match the Mercatura network identity expected by this MercaMiner build. Check the selected network and make sure Mercatura Core and MercaMiner are from compatible revisions.

Signet does not mine

This is intentional in v0.1. Signet identity is recognized, but direct PoW mining is blocked.

Hashrate looks low immediately after startup

Give the average time to converge. On the validated development system, the displayed rate typically became much more stable after a few minutes.

CPU temperature is too high

Reduce the worker count and benchmark again. A lower worker count may lose little hashrate while substantially reducing power, memory pressure, and temperature.

16. Quick operational checklist

- ☐ Mercatura Core is running on the intended network.
- ☐ The node reports the expected chain and is usable.
- ☐ A payout address for that same network is ready.
- ☐ MercaMiner builds successfully and its tests pass.
- ☐ The selected worker count fits available RAM and cooling.
- ☐ The miner is using either automatic local RPC discovery or a correct `rpc_url/cookie_file` pair.
- ☐ The first status lines show the expected network, workers, and height.
- ☐ Accepted increases when blocks are successfully mined; rejected should remain zero during normal operation.
- ☐ Use Ctrl+C for a clean shutdown.

Appendix A - Tested continuous testnet example

The following form matches the successful local testnet validation pattern used during v0.1 development. Replace the address and paths if using a different testnet instance.

```
cd ~/projects/mercaminer
./build/mercaminer \
  --network test \
  --payout-address YOUR_TESTNET_MERCATURA_ADDRESS \
  --threads 6 \
  --block-limit 0 \
  --report-interval 30 \
  --rpc-url http://127.0.0.1:27775 \
  --cookie-file /YOUR/DATADIR/testnet/.cookie
```

The validated live session reached 50 accepted blocks with 0 stale and 0 rejected blocks and shut down cleanly with Ctrl+C.

Appendix B - v0.1 scope

Included in MercaMiner v0.1:

- Linux x86_64 CPU mining
- solo JSON-RPC mining
- mainnet/testnet/regtest direct PoW support
- multi-threaded mining
- configuration-file and CLI operation
- automatic local RPC discovery
- runtime statistics
- stale/rejected classification
- benchmark tooling

Not included in v0.1:

- GPU mining
- Stratum or mining-pool protocols
- direct signet mining
- Bitcoin Testnet4
- graphical user interface